Certificate-Gated Dynamic Routing under Compound Urban Disruptions

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1. Problem Statement

Road conditions in Chennai can change during monsoon flooding, incidents, and congestion. A road may remain physically connected while losing speed or capacity; redirected vehicles may then overload the remaining alternatives. Recomputing every route after every small update is expensive, but using stale road costs for too long can produce poor routes.

The research problem is:

> How can a routing system update many Chennai vehicle routes under changing flood, incident, and traffic conditions while controlling computation, route error, unnecessary switching, and congestion created by the recommendations themselves?

The system uses a fixed directed road graph with edge costs that change between routing epochs. This is **snapshot-dynamic routing**, not formal time-dependent routing unless edge cost becomes a function of edge-entry time inside one query.

2. Objective

Develop and evaluate a reproducible framework that:

1. maps Chennai road, flood, rainfall, incident, and traffic evidence to explainable road states;
2. represents disruption through road availability and effective capacity;
3. converts flow and capacity into travel time using the BPR model;
4. uses Customizable Contraction Hierarchies (CCH) for repeated shortest-path queries;
5. avoids unnecessary CCH metric refreshes with a bounded-staleness certificate;
6. controls route changes using minimum-degradation, minimum-gain, and cooldown rules;
7. includes projected rerouting demand before assigning further routes;

8. evaluates critical-facility access, population-weighted loss, partial compliance, and an emergency scenario; and
9. compares all outcomes with Dijkstra and other justified baselines.

3. Expected Outcome

The expected outcome is an evaluated research prototype—not an operational navigation service. It should establish:

- whether certificate-gated synchronization reduces CCH refresh work without violating its declared route-quality bound;
- when CCH is preferable to repeated Dijkstra for the observed query/update workload;
- whether stability and projected-load controls reduce route churn and detour congestion;
- how compound disruption changes access to hospitals, fire stations, and relief centres;
- how benefits change under incomplete guidance compliance and uncertain data; and
- which conclusions are supported only by synthetic experiments versus Chennai evidence.

4. Research Gap and Contribution

4.1 Novelty Verdict

The project does **not** introduce a new shortest-path algorithm or a new approximation-certificate principle.

The following are established:

- Dijkstra, A*, ALT, CCH, CATCHUp, LPA*, and D* Lite;
- BPR travel-time functions and capacity reduction;
- flood-aware routing and flood–SUMO coupling;
- dynamic rerouting thresholds, bounded rationality, and cooldown;
- projected-load or route-reservation routing;
- emergency priority;
- lower-bound/upper-bound certificates for bounded-suboptimal paths.

The strongest defensible contribution is:

> A reproducible Chennai-oriented compound-disruption framework that applies an established monotone lower/upper-bound certificate as a query-level CCH refresh gate, and is designed to experimentally separate routing-index performance from the traffic effects of stable, projected-load-aware route adoption.

Provisional novelty assessment: Medium for the proposed integration/evaluation and Low for algorithmic novelty, pending a systematic database search and completed Chennai experiments.

4.2 Closest Prior Art

Existing work	What already exists	Remaining difference
Bono et al., CPD-Search, IJCAI 2019	Old shortest distance as a lower bound, old path re-evaluated as an upper bound under increasing costs	Uses a compressed path database and A* repair, not a CCH refresh gate
Aine and Likhachev, 2016	Truncated incremental repair with bounded suboptimality	Query-specific incremental search, not batched CCH customization
Dibbelt et al., CCH, 2016	Metric-independent preprocessing, customization, fast exact queries, partial propagation	No per-query certificate for deferring refresh
Buchhold et al., 2019	CCH with BPR traffic assignment and batched queries	No flood state, stability controller, or certificate gate
Chan et al., 2023	CCH customization/query costs, recheck periods, improvement thresholds, compliance/penetration, and congestion redistribution	No flood evidence or deterministic monotone per-query stale-metric certificate
CERT-FLOW, 2026 preprint	Implemented proof-gated CH/oracle routing under drifting costs	Probabilistic conformal bounds and dual search, not the deterministic monotone specialization
Li et al., 2026	Hydrodynamic flooding, SUMO, rerouting, and emergency vehicles	No CCH certificate or explicit projected-load/stability evaluation
Pan et al., 2012	Proactive projected vehicle footprints and sequential rerouting	No flood evidence or CCH

No “first certificate,” “first CCH traffic assignment,” or “first flood-aware Chennai router” claim is supportable.

Search Protocol and Limit

The audit used targeted English-language searches through 6 September 2026 across scholarly web indexes, DOI/publisher records, surveys, reference chains, and Google Patents. Search concepts included `CCH dynamic rerouting`, `lazy/partial customization`, `stale metric shortest path certificate`, `monotone edge-weight increase`, `bounded suboptimal replanning`, `BPR route reservation`, `flood SUMO routing`, and `Chennai flood routing`. Candidate work was screened for certificate logic, routing index, traffic assignment, flood evidence, stability, and emergency/accessibility evaluation.

This was a focused prior-art audit, not a registered systematic review. The claimed gap is therefore provisional and must be rechecked using venue-specific databases and documented inclusion/exclusion counts before submission.

4.3 Chennai-Specific Gap

Chennai work already includes:

- crowdsourced flooded-street mapping (Naik, 2016);
- flood-relief vehicle routing (Ganguly and Roy, 2017);
- city-level Dijkstra/A*/ALT routing (Kumar et al., 2020);
- flood forecasting through C-FLOWS (publisher PDF);
- flood susceptibility mapping (Alabdan et al., 2025);
- SUMO calibration for heterogeneous Chennai traffic (Sashank et al., 2020);
- Chennai BPR-family calibration (Gore, Arkatkar, Joshi, and Antoniou, 2023); and
- a recent flood/traffic/safety navigation concept (2026 paper).

No verified Chennai study was found that jointly evaluates flood-dependent effective capacity, CCH refresh behavior, projected route load, route stability, partial compliance, and population-weighted critical-facility access. This is an evidence-based search result, not proof of universal absence.

4.4 Publication Positioning

The paper should be positioned as an **integration, systems, and experimental evaluation paper**. The certificate is an established principle specialized to CCH synchronization. Publication strength depends on:

- transparent Chennai data provenance;
- calibrated or sensitivity-tested capacity assumptions;
- native CCH versus Dijkstra workload benchmarks;
- SUMO outcome evaluation;
- ablations isolating the certificate, CCH, stability, projected load, and priority;
- public experiment configurations and negative results.

4.5 Historical Data and Publication Validity

The absence of a public live Chennai traffic/closure feed does not by itself preclude a retrospective methods submission; suitability depends on validation quality and venue. The study must be framed as a **retrospective historical-evidence-conditioned simulation/scenario reconstruction**, not an observed road-state replay or operational live deployment.

Historical evidence can strengthen reproducibility because every method is evaluated against the same dated event. The evaluation will:

1. condition scenarios on documented Chennai flood evidence and rainfall for declared dates;
2. preserve source versions, retrieval dates, checksums, and spatial resolution;
3. separate model calibration/sensitivity from held-out scenario evaluation;
4. inject controlled 0/30/60/120-minute lags and false-positive/false-negative states into derived/simulated road-state traces;
5. use repeated SUMO seeds and report confidence intervals;
6. compare historical-evidence routing with perfect-information and no-flood baselines; and
7. describe the architecture as **designed for near-real-time operation**, not as a validated near-real-time or live Chennai service.

OpenCity hotspot/hazard layers do not provide authoritative timestamped road-state trajectories. Until such observations are obtained, lag/error tests measure robustness to controlled assumptions rather than empirical sensing accuracy. The study cannot prove present-day live accuracy or deployment readiness.

5. Proposed Method

5.1 Network and Cost Model

Let ($G=(V,E)$) be a fixed directed multigraph. Every edge has:

- free-flow time (t^0_e);
- baseline capacity (c^0_e);
- assigned entering demand ($x_{e,t}$);
- flood multiplier ($m^{flood}_{e,t}$);
- incident multiplier ($m^{incident}_{e,t}$); and
- source timestamp/confidence.

Effective capacity is:

[$c^{eff}_{e,t}=$

]

The BPR route cost is:

[t_{e,t}=t^0_e,]

where adjacency denotes multiplication. For software and CCH, seconds are quantized to non-negative integer milliseconds. Flow and capacity use the same interval and units. Discharged throughput is retained as an outcome, not substituted for assigned entering demand.

When (c^{eff}_{e,t}=0), the edge is excluded and its routing cost is (+); the finite BPR expression is not evaluated. Native CCH currently rejects (+) closures; Dijkstra omits those edges. An integer-millisecond conversion is implemented for the Stage 2 snapshot glue.

5.2 Which Algorithm Finds the Path?

Stage 1 uses Dijkstra. Stage 2 can select Dijkstra or the native CCH adapter on synthetic graphs. The planned Chennai implementation will use CCH only after Stage 6 validates topology conversion, turns, closures, quantization, ordering, and path equality.

- **Dijkstra:** implemented correctness oracle and Stage 1 baseline.
- **CCH:** implemented prototype adapter; planned final Chennai path engine after validation.
- **ALT-guided bidirectional A*:** unimplemented planned comparator/backup.
- **Certificate gate:** decides whether CCH must be refreshed; it is not a path-finding replacement.

The repository now contains:

- `RoutingKitCCHEngine`: native experimental CCH adapter;
- `NetworkXDijkstraEngine`: exact reference adapter;
- `CertifiedLazySynchronizer`: certificate and refresh controller;
- `EagerRefreshRouter`: repeated-refresh baseline.

5.3 Certified Lazy Synchronization

Let:

- (w): the edge metric currently synchronized into CCH;
- (w): the latest authoritative/projected metric;
- (P_0): an exact path returned under (w);
- (L=d_{w}(s,t)): its old optimal distance;
- (U=C_w(P_0)): the same keyed-edge path evaluated under current weights; and
- (): permitted represented-cost stretch.

When every current edge weight is at least its synchronized value:

[w_ew_e e,]

the old optimum is a current lower bound. The route can be returned without CCH customization when:

[U(1+)L.]

Otherwise, the controller customizes CCH with the complete current metric and queries again.

Three-Road Example

Assume three parallel roads:

Road	Synchronized cost
A	5.0 min
B	6.0 min
C	8.0 min

CCH selects Road A and records (L=5.0).

With (%):

- projected demand raises A to 5.2 min;
- allowed cost is (1.05=5.25);
- (5.2), so A is certified and CCH is not refreshed.

If A rises to 5.8 min:

- (5.8>5.25);
- the certificate fails;
- CCH is customized with A=5.8, B=6.0, C=8.0 and queries again.

If A rises to 7.0 min, refreshed CCH selects B at 6.0 min.

5.4 Formal Guarantee

Assumptions: fixed topology, complete atomic metric snapshots, non-negative integer represented weights, exact synchronized-engine query, and pointwise nondecreasing current weights.

For every path (P), (C_{w}(P)C_w(P)). Therefore:

[L=d_{w}(s,t)d_w(s,t).]

Because (P_0) remains feasible:

[d_w(s,t)U=C_w(P_0).]

If (U(1+)L), then:

[U(1+)L(1+)d_w(s,t).]

Thus the returned path is within (1+) of the current represented optimum.

This is an application of an established upper/lower-bound certificate, not a new theorem.

5.5 Mandatory Refresh Conditions

The lower bound is invalid if any current weight drops below the synchronized weight. The implementation refreshes before querying after:

- flood or incident recovery;
- capacity increase;
- road reopening;
- any other represented cost decrease.

A closure represented as an increase is safe for the lower-bound direction, but a candidate containing the closed edge fails the upper-bound test. Native CCH currently accepts finite integer weights only; closure-sentinel behavior still requires a dedicated validation stage.

5.6 Route Adoption and Projected Load

The certificate controls **metric synchronization**. A separate SCENARIO filter can then decide **whether a vehicle changes route**:

1. current route becomes infeasible, or degradation exceeds (_{deg});
2. candidate improvement exceeds (_{gain});
3. cooldown has expired unless safety requires immediate action.

Items 1–3 are implemented as `decide_route_adoption`. Time-indexed reservations (items previously 4–5) remain unimplemented. The filter is not claimed to achieve traffic equilibrium.

5.7 Emergency and Public-Service Evaluation

Emergency routing remains a secondary scenario:

- blocked roads remain forbidden;
- emergency deadline/arrival time receives priority;
- delay imposed on ordinary traffic is reported;
- no signal preemption or live ambulance tracking is claimed.

Five committed evaluation studies strengthen impact without becoming route weights:

1. **Evidence freshness and uncertainty:** controlled lag and classification-error traces.
2. **Critical-facility accessibility:** travel time/disconnection to hospitals, fire stations, and relief centres.
3. **Population-weighted access loss:** distribution of access impact using WorldPop/ward weights; this is not socioeconomic equity.
4. **Partial compliance:** nearest-integer seeded cohorts target 0%, 25%, 50%, 75%, and 100%; exact percentages require a compatible fleet size.
5. **Facility-oriented criticality:** rank directed keyed arcs by newly disconnected population and unnormalised person-time added to facility access; group both directions/parallel arcs by OSM way ID before physical-road reporting.

Generic utility functions exist for portions of all five factors in `evaluation/metrics.py` and `evaluation/robustness.py`. None of the five has yet been integrated with Chennai inputs or executed as a factor-level experiment.

6. Dataset and Data Access

6.1 Factor-to-Data, Credential, and Implementation Matrix

Factor	Required data	Chennai source	Access method	API key/ac
Evidence freshness and uncertainty	A derived/simulated binary road-state truth trace plus rainfall/flood evidence	OpenCity flood records; Open-Meteo reanalysis; optional IMERG Final V07	CKAN download, HTTPS JSON, or credentialed GES DISC download	**No key** for OpenCity/C Meteo; Ear account authorizati plus credentials for optiona IMERG
Critical-facility accessibility	Geolocated health, fire, and relief facilities	OpenCity health, fire, and relief; OSM POIs	Health CSV/selected KML, fire CSV/KML, relief-centre PDF, and cached Overpass response	**No key or account** fo listed sourc
Population impact	Population count raster	Exact 2015 India 1 km R2025A STAC item	Public STAC asset/GeoTIFF download	**No key or account**
Partial compliance	Vehicle IDs and declared compliance level; no observed	SUMO automatic-routing	Local SUMO configuration and	**No key or account**

	compliance dataset	options	TraCI/libsumo	
Facility-oriented criticality	Chennai graph + facilities + population; no separate dataset	Derived from OSM, OpenCity facilities, and WorldPop	Local reverse multi-source shortest-path analysis	**No additio key or acc**

Feasibility conclusion: a public-source acquisition path exists for all five factors without user-supplied credentials, but the complete Chennai pipeline has not yet been demonstrated. Official NASA IMERG/SRTM data cannot be acquired in a no-credential run; Open-Meteo reanalysis and non-elevation susceptibility inputs are the declared fallbacks.

6.2 API and Authentication Checklist

Source/service	Cost for research core	Key required?	Account required?	Access/auther detail
OpenCity Chennai	Free public data	No	No	Dataset page ar resource down
Public OSM Overpass	Free community service	No	No	`https://overpa api.de/api/int` use caching and descriptive use
Geofabrik India OSM extract	Free public download	No	No	Download date `.osm.pbf`; a sep importer is req
Open-Meteo Historical Weather API	Free for rate-limited non-commercial use	No	No	`https://archiv api.open-meteo.com/v1/a` returns JSON
NASA GPM IMERG/GES	Free	No conventional	**Yes**	Authorize NAS DISC and use E credentials or

IMERG/GES DISC	Free	conventional key	Yes	credentials or `Authorization: <token>`
WorldPop India 2015 1 km STAC item	Free public download	No	No	Select the GeoT from the fixed i do not discover `datetime` alone
Eclipse SUMO/TraCI	Free open-source software	No	No	Local installati Python/TraCI ir
RoutingKit CCH	Free open-source package	No	No	Install `routingk` Python package

No secret, password, or bearer token is committed to the repository.

6.3 OpenStreetMap

Purpose: Directed Chennai road topology and road attributes.
Contains: Nodes, ways, geometry, road class, direction, and incomplete lanes/speeds/turn data.
Chennai coverage: Yes; completeness must be audited.
Global coverage: Yes.
Access: OSMnx/Overpass or dated Geofabrik extract.

Limitations: Mutable community data; missing attributes and turn restrictions.
Direct access: Geofabrik India
Documentation: OSM licence, Overpass API policy
Repository: OSMnx
Viewer: Chennai map

The facility query will be sent to `https://overpass-api.de/api/interpreter`, cached with retrieval time/checksum, and limited to the declared Chennai study boundary. A descriptive User-Agent is mandatory. The research job will use one non-parallel request, avoid large repeated queries, and treat the public instance as best-effort with no SLA.

```
[out:json][timeout:120];
(
  nwr["amenity"~"^(hospital|clinic|fire_station)$"]
(12.85,80.05,13.25,80.40);
  nwr["social_facility"="shelter"](12.85,80.05,13.25,80.40);
);
out center tags;
```

The bounding box is a provisional acquisition envelope and must be replaced by the final Stage 3 study boundary. `social_facility=shelter` is exploratory OSM tagging and must not be equated with an activated GCC relief centre. The official GCC PDF remains authoritative for the declared 2024 list.

Relief-address geocoding will follow the Nominatim usage policy: maximum one request/second, one thread on one machine, valid identifying User-Agent, local caching, attribution, and no recurring bulk job. Coordinates/names will be manually verified; unverified entries will be excluded.

6.4 OpenCity Chennai Flood Data

Purpose: Historical flood evidence and susceptibility validation.
Contains: The 2015 dataset has hotspots/stagnation points and a 2015 inundation zone; the separate Chennai Flooding Data collection has inundation points/depth and return-period hazards.
Chennai coverage: Direct.
Global coverage: No.
Access: Public CKAN resource download in KML.

Limitations: Historical/modelled evidence is not a current road closure.
Direct access: Chennai Floods 2015, Chennai Flooding Data
Documentation: CKAN API metadata
Repository: `flood.py`
Viewer: Resource previews on the OpenCity page.

6.5 NASA GPM IMERG

Purpose: Historical and delayed near-current rainfall forcing.
Contains: Half-hourly satellite precipitation; historical evaluation selects Final V07 collection `GPM_3IMERGHH_07`.
Chennai coverage: Yes, at approximately 0.1° cells.
Global coverage: Near-global.
Access: Free Earthdata account, NASA GES DISC application authorization, and `.netrc` credentials or bearer-token authentication.

Limitations: Approximately 10 km cells; rainfall does not prove street flooding.
Direct access: GPM_3IMERGHH_07 summary
Documentation: IMERG V07, DOI 10.5067/GPM/IMERG/3B-HH/07
Repository: `src/chennai_routing/data/rainfall.py` is currently a placeholder; acquisition is unimplemented.
Viewer: NASA Giovanni

If no Earthdata credentials are available, the no-key Open-Meteo Historical Weather API provides an ERA5-derived fallback. A tested Chennai request uses `latitude=13.0827`, `longitude=80.2707`, declared dates, `hourly=precipitation`, `models=era5`, and `timezone=Asia/Kolkata`. It must be labelled retrospective reanalysis—not contemporaneous sensing or road-level observation.

6.6 NASA SRTM/NASADEM and Chennai Hydrology

Purpose: Static flood-susceptibility context.
Contains: Approximately 30 m elevation plus separate drain, canal, river, and water-body vectors.
Chennai coverage: Yes.
Global coverage: Terrain is broad/global; OpenCity hydrology is Chennai-specific.
Access: Free Earthdata login and public OpenCity downloads.

Limitations: Terrain is not road-level flood depth; drain maps do not prove capacity or maintenance.
Direct access: SRTMGL1
Documentation: OpenCity drains
Repository: `src/chennai_routing/data/elevation.py` and `hydrology.py` are currently placeholders.
Viewer: Earthdata Search

6.7 Eclipse SUMO

Purpose: Dynamic traffic, queues, incidents, vehicle classes, and realized outcomes.
Contains: Software-generated vehicle/edge simulation output—not observed Chennai traffic.
Chennai coverage: User-constructed from the project graph and demand.
Global coverage: User-defined.
Access: Free open-source installation and TraCI/libsumo.

Limitations: Requires Chennai demand/behaviour calibration.
Direct access: SUMO
Documentation: SUMO documentation, automatic-routing/compliance options
Repository: Eclipse SUMO
Viewer: SUMO-GUI, not a web data viewer.

6.8 Critical Facilities and Population

Purpose: Evaluate public-service accessibility and population-weighted impact.
Contains: Health-centre CSV/selected KML, fire-station CSV/KML, a 2024 relief-centre PDF, and modelled population counts.
Chennai coverage: Direct for OpenCity facilities; WorldPop covers Chennai.
Global coverage: WorldPop is global; facility catalogues are local.
Access: Public OpenCity downloads and the fixed WorldPop STAC item/GeoTIFF asset.

Limitations: UPHC/UCHC data is not a comprehensive emergency-hospital inventory; the relief PDF does not provide validated graph-ready coordinates or current activation; population exposure is not socioeconomic equity.
Direct access: Health metadata/API, fire metadata/API, relief metadata/API, WorldPop 2015 India 1 km item, 15.48 MB GeoTIFF asset
Documentation: WorldPop 1 km dataset DOI, WorldPop STAC
Repository: `src/chennai_routing/evaluation/metrics.py` provides array/graph utilities; ingestion and graph snapping are unimplemented.
Viewer: OpenCity resource previews and WorldPop STAC Browser.

Selected OpenCity resources were downloaded and verified on 6 September 2026:

Selected resource	CKAN resource ID	Format/size	SH
UPHC map	`5a81427f-0aa1-4270-8df4-5d4cef250912`	KML, 100,497 B	`a700d7e837a5466d38e99a6ed1b67a`
UCHC map	`cd4effee-997f-4fe0-b376-8801080c6962`	KML, 14,155 B	`45e31b7dc026f54dd4e03191c2b912`
Fire-station locations	`39d3601b-cd42-4c11-a6c3-8fc4b057b38a`	KML, 13,664 B	`f41a5afb44e316aea26403f8403b08`
GCC 2024 relief-centre list	`ee9f087a-dc6c-41fa-8810-954a9f5887ac`	PDF, 247,507 B	`9d68bb27098d81b161badb721a12d7`

These resources identify primary/community health centres, fire stations, and a relief-centre list; they do not establish trauma capability, facility capacity, opening status, or emergency readiness.

7. Data Classification and Temporal Meaning

Source	Classification	Historical/current meaning
OSM	Primary input	Mutable map snapshot
OpenCity flood	Historical evidence	Past observation/modelled hazard
SRTM/NASADEM	Supporting input	Static 2000-era terrain

Open-Meteo ERA5	No-key core rainfall input	Retrospective reanalysis, not contemporaneous sensing
IMERG Final	Optional credentialed calibration input	Historical gauge-adjusted satellite rainfall
IMERG Early	Optional credentialed near-current input	Delayed satellite rainfall estimate
SUMO	Experimental generator	Simulated traffic
Facility data	Evaluation input	Catalogue snapshot
WorldPop	Evaluation input	Modelled population
Processed edge metric	Derived dataset	Project calculation with version/time

Optional forecasts may support demonstrations, but the historical core uses no-key Open-Meteo ERA5 reanalysis and does not depend on paid APIs. No verified public live Chennai road-speed, accident, signal, or closure API is assumed.

8. Data-to-Routing Mapping

Raw source	Derived factor	Model effect	Routing effect
OSM	Topology, free-flow time, capacity assumption	Base graph	Feasible paths/lower cost
Flood history + terrain + drains	Susceptibility	Static prior	Modifies rainfall response
Open-Meteo ERA5 or optional IMERG	Rolling rainfall	Scenario-conditioning evidence	Capacity/availability assumption
Verified closure/incident	Edge state	Capacity zero/reduction	Remove/raise edge cost
Assigned SUMO demand	PCE/time entering flow	BPR (x/c)	Metric update
Accepted compliant routes	Projected edge-entry load	Future BPR metric	Reduces herding
Facilities + population	Access outcomes	Evaluation only	No arbitrary route penalty

9. Architecture

```
flowchart TD
    Data[Road_Flood_Rain_Incident_Data] -->
State[Explained_Road_State]
    State --> Capacity[Effective_Capacity]
    Demand[SUMO_and_Projected_Demand] --> BPR[BPR_Integer_Metric]
    Capacity --> BPR
    BPR --> Monotonicity[Check_Current_Weights_ge_Synchronized]
    OD[Origin_Destination_Query] --> Monotonicity
    Monotonicity -->|Nondecrease| StaleQuery[Query_Synchronized_CCH]
    Monotonicity -->|Any_Decrease|
Customize[Customize_CCH_with_Current_Metric]
    StaleQuery --> PathEval[Evaluate_Old_Path_on_Current_Metric]
    BPR --> PathEval
    PathEval --> Gate[LB_UB_Certificate_Gate]
    Gate -->|Pass| Candidate[Certified_Candidate_Route]
    Gate -->|Certificate_Fail| Customize
    BPR --> Customize
    Customize --> FreshQuery[Query_Refreshed_CCH]
    FreshQuery --> Candidate
    Candidate --> Policy[Stability_Priority_Compliance]
    Policy --> Reservation[Projected_Load_Reservation]
    Reservation --> Demand
    Policy --> SUMO[SUMO_Realized_Traffic]
    SUMO --> Evaluation[Travel_Stability_Access_Evaluation]
    Dijkstra[Dijkstra_Oracle] --> Evaluation
```

Implemented now: integer BPR snapshot glue, certificate gate, Dijkstra/CCH engines, and a SCENARIO adoption-threshold filter. SUMO, projected reservations, and Chennai road-state ingestion remain unimplemented on this branch.

10. Implementation Status and Revised Stages

Stage 1 — Previously Executed Historical-Hotspot-Seeded Controlled Demo

Status: Previously executed; implementation remains, but publication-grade evidence is not versioned.
OSM and OpenCity historical hotspots were joined to roads; one real mapped edge was controlled as blocked; BPR weights and two Dijkstra snapshots demonstrated closure avoidance. Inputs/outputs are ignored and current tests use fixtures. This becomes publication evidence only after dated inputs, checksums, configuration, and a compact result manifest are preserved. Finite congestion response is not validated.

Stage 2 — Certificate and Routing-Engine Validation

Status: Implemented as a synthetic preliminary experiment.

- engine-neutral complete metric snapshots;

- exact NetworkX Dijkstra engine;
- native `routingkit-cch` 0.1.4 engine;
- keyed-edge path extraction;
- integer metric and atomic updates;
- bounded-staleness certificate;
- mandatory refresh after lower-bound invalidation;
- eager baseline and deterministic experiment runner;
- accessibility/compliance evaluation utilities;
- uncertainty and facility-road-criticality utilities;
- 55 collected tests when `routingkit-cch` is installed (4 skip without it).

Stage 3 — Reproducible Chennai Graph

Status: Planned.
Dated OSM extract, stable arc IDs, validated directions/turns/parallel arcs, free-flow times, and documented capacity assumptions.

Stage 4 — Flood and Road-State Evidence

Status: Synthetic binary lag/error utility implemented; dated Chennai road-state truth and temporal integration are unimplemented.
Historical susceptibility, IMERG or no-key reanalysis rainfall, optional current evidence, source freshness, confidence, and explained `NORMAL/DEGRADED/SEVERE/BLOCKED` states.

Stage 5 — Chennai Traffic and SUMO Calibration

Status: Planned.
Heterogeneous demand, entering PCE flow, incidents, queues, BPR calibration/sensitivity, and no double counting of SUMO delay.

Stage 6 — Chennai CCH Integration

Status: Planned.
Geometry-aware ordering, turn-expanded topology, finite closure sentinel, integer quantization, full/partial customization, and Dijkstra differential validation.

Stage 7 — Stable Projected-Load Rerouting

Status: SCENARIO threshold/cooldown adoption filter implemented; cooldown does not delay degradation or infeasible incumbents. Time-indexed reservations and SUMO comparison are unimplemented.

Stage 8 — Accessibility, Population, Compliance, and Road Criticality

Status: Generic utility portions implemented; no Chennai factor-level experiment has run.
Hospitals, fire stations, relief centres, disconnection, p90 access, population-weighted loss, partial compliance, and facility-oriented road criticality.

Stage 9 — Emergency Scenario

Status: Planned.
Response time, deadline success, safety exposure, and ordinary-user external delay.

Stage 10 — Full Evaluation and Paper

Status: Preliminary method evidence available; Chennai/SUMO evidence pending.
Paired scenarios, baselines, ablations, uncertainty, statistical reporting, reproducibility package, and final manuscript.

11. Preliminary Certificate Experiment

11.1 Setup

- seeded synthetic directed multigraph;
- 200 nodes plus 600 additional arcs;
- 100 update epochs;
- 5 edge updates and 50 OD queries per epoch;
- 5% certificate tolerance;
- identical trace for Dijkstra and CCH;
- monotone-increase and mixed increase/decrease workloads (a changed edge decreases with probability 0.3);
- Python 3.12.3, NetworkX 3.6.1, `routingkit-cch` 0.1.4.

11.2 Main Results

Prototype adapter/workload	Queries	Certified stale	Lazy refreshes	Eager update refreshes	Avoide
Dijkstra/increases	5,000	4,840	6	100	9
Dijkstra/mixed	5,000	700	86	100	1
CCH/increases	5,000	4,840	6	100	9
CCH/mixed	5,000	700	86	100	1

There were zero certificate violations, zero exact post-refresh mismatches, and zero CCH-versus-independent-Dijkstra oracle mismatches. The monotone workload produced the largest reduction because the lower bound remained valid. Mixed decreases correctly forced frequent refreshes; in this single run, lazy CCH was slightly slower than eager CCH.

These are one-machine, single-run synthetic prototype-adapter measurements with degree ordering. “Total” uses symmetric wall-clock boundaries for all post-initialization updates and route requests. Engine construction is reported separately; synchronization diagnostics combine initial and later synchronizations and should not be added to “Total.”

The CCH adapter constructs a fresh `CCHMetric` on synchronization and a fresh `CCHQuery` for each request; metric reset, partial/parallel customization, and query reuse are not benchmarked. The eager Dijkstra adapter similarly rebuilds its represented weighted graph. Therefore, these values validate prototype behavior only and do not establish an optimized CCH-versus-Dijkstra break-even point or Chennai performance.

11.3 Epsilon Sensitivity

For 2,500 CCH queries over 50 monotone epochs, refreshes fell from 33 at 0% tolerance to 25 at 1%, 4 at 5%, and 0 at 10%, with zero observed bound violations. Under mixed updates, decreases dominated: refreshes were 47, 44, 43, and 43 respectively.

Machine-readable results are stored in `docs/evidence/CERTIFIED_LAZY_SYNC_RESULTS.json`.

12. Evaluation Plan

Baselines

1. route-once Dijkstra;
2. eager repeated Dijkstra;
3. eager CCH;
4. certificate-gated CCH;
5. ALT-guided bidirectional A*;
6. SUMO periodic rerouting;
7. independent versus projected-load-aware assignments.

Ablations

- no flood evidence;
- no certificate;
- no threshold/cooldown;
- no projected reservations;
- no emergency priority;
- no population weighting;

- compliance levels 0–100%.

Metrics

Metric	Purpose
Certificate violation and optimality gap	Method correctness
Preprocess/customize/query/unpack/epoch latency	Routing feasibility
Travel time, person-delay, queue, spillback	Traffic outcomes
Reroute count and reversals	Stability
Maximum (x/c) and load concentration	Herding
Blocked/severe-edge exposure	Safety
Facility travel-time p50/p90 and disconnection	Public-service access
Population-weighted access loss	Distributional impact
Emergency deadline success and normal-user delay	Priority trade-off
Lag/error sensitivity	Data uncertainty

13. Reproducibility

```
python -m venv .venv
source .venv/bin/activate
python -m pip install -e ".[test,cch]"
python -m pytest
python scripts/run_certified_lazy_experiment.py \
  --engine both --mode both --seed 8597 \
  --nodes 200 --extra-edges 600 \
  --epochs 100 --updates-per-epoch 5 \
  --queries-per-epoch 50 --epsilon-percent 5
```

Stage 1:

```
python scripts/run_stage1_poc.py
```

Remote OSM/OpenCity inputs remain mutable until dated snapshots and checksums are committed. Synthetic experiment results are deterministic apart from timing.

14. Feasibility and Remaining Uncertainty

Public/no-key inputs and tools: OSM, OpenCity, Open-Meteo reanalysis, facility catalogues, WorldPop, SUMO, NetworkX, native CCH binding, and the tested Python utilities. **Credentialed optional inputs:** official IMERG and SRTM/NASADEM through Earthdata.

Uncertain/optional: public live Chennai speeds, machine-readable closures/incidents, signal-controller data, ambulance AVL, street-level satellite flood depth, and facility capacity.

The proposed core remains feasible as historical-evidence-conditioned scenario reconstruction and labelled simulation without paid APIs; end-to-end acquisition and Chennai execution are not yet demonstrated.

15. Limitations

1. The certificate principle has close prior art; algorithmic novelty is not claimed.
2. Preliminary experiments are synthetic and do not establish Chennai outcomes.
3. Native CCH uses degree ordering; production Chennai ordering is unvalidated.
4. CCH closure sentinel, turns, and quantization require further tests.
5. Any weight decrease invalidates the stale lower bound and usually forces refresh.
6. The certificate controls represented route cost, not model accuracy.
7. BPR parameters and flood-capacity multipliers are not Chennai-calibrated.
8. BPR does not represent queue spillback; SUMO must measure it.
9. Historical flood observations are not live closures.
10. IMERG is coarse and delayed relative to streets.
11. SUMO demand is simulated and requires local calibration.
12. Population exposure is not socioeconomic equity.
13. Facility catalogues do not provide capacity or guaranteed emergency capability.
14. Partial compliance is a sensitivity assumption, not observed behaviour.
15. Projected greedy reservations do not guarantee equilibrium.
16. A literature audit cannot prove universal novelty.
17. All five strengthening factors currently have only generic utility portions, not completed Chennai experiments.
18. OpenCity historical layers do not provide authoritative timestamped road-state truth.
19. Stage 1 uniform flow/capacity makes BPR almost a scale factor; congestion response is not validated.
20. The adoption filter is a labelled policy, not observed driver behaviour or an equilibrium model.

16. Final Project Summary

Component	Final decision
Problem	Repeated routing under compound flood, incident, and congestion updates
	Dijkstra implemented; prototype

Path engine	CCH implemented; Chennai CCH planned after Stage 6; ALT unimplemented comparator
Certificate	Established LB/UB principle used as a CCH refresh gate
Defensible novelty	Chennai integration, workload characterization, and compound-disruption evaluation
Flood model	Susceptibility + rainfall + optional observation → explained state
Traffic model	BPR route estimate + SUMO realized outcome
Stability	Trigger, minimum gain, cooldown, projected time-indexed load
Additional factors	Uncertainty, facility access, population-weighted loss, partial compliance, road criticality
Emergency	Secondary scenario with external-delay reporting
Current evidence	Tested synthetic CCH/Dijkstra method experiment; Stage 1 demo was previously run but lacks versioned publication evidence
Required next evidence	Chennai graph, flood/rain pipeline, SUMO calibration and full ablations

17. Research Claim Boundary

The completed implementation supports this statement:

> On deterministic synthetic fixed-topology workloads, the implemented monotone metric certificate returned routes within its declared represented-cost bound, matched exact routing after refresh, and reduced eager metric refreshes for both Dijkstra and CCH adapters.

It does **not yet** support:

- improved Chennai travel time or emergency response;
- operational real-time flood routing;
- a new shortest-path algorithm;
- a new approximation-certificate theorem;
- city-scale CCH performance;
- publication-ready causal claims.

Those claims require completion of Stages 3–10.